

SCENARIO

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NAME	MATRIC NO	GROUP
IRDINA NADHIRAH BINTI BAHANUDDIN	22C18G026	2

PROBLEM:

Write the C++ program that accepts a student's numerical grade, converts the numerical grade to an equivalent letter grade, and displays the letter grade. Start with a pseudocode and flowchart.

SAMPLE OUTPUT:

Sample Output:

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IPO CHART		
INPUT	PROCESS	OUTPUT
Numerical_grade	if (Numerical_grade >= 90 && Numerical_grade <= 100) Print_Grade_A; else if (Numerical_grade >= 80 && Numerical_grade <= 89) Print_Grade_B; else if (Numerical_grade >= 70 && Numerical_grade <= 79) Print_Grade_C; else if (Numerical_grade >= 60 && Numerical_grade <= 69) Print_Grade_D; else if (Numerical_grade >= 0 && Numerical_grade <= 59) Print_Grade_F; else Print_Invalid_input;	Print_Grade

PSEUDOCODE:		

Start

Prompt Numerical_grade

Read Numerical_grade

if (Numerical_grade>=90 && Numerical_grade<=100)

 Print_Grade_A;

else if (Numerical_grade>=80 && Numerical_grade<=89)

 Print_Grade_B;

else if (Numerical_grade>=70 && Numerical_grade<=79)

 Print_Grade_C;

else if (Numerical_grade>=60 && Numerical_grade<=69)

 Print_Grade_D;

else if (Numerical_grade>=0 && Numerical_grade<=59)

 Print_Grade_F;

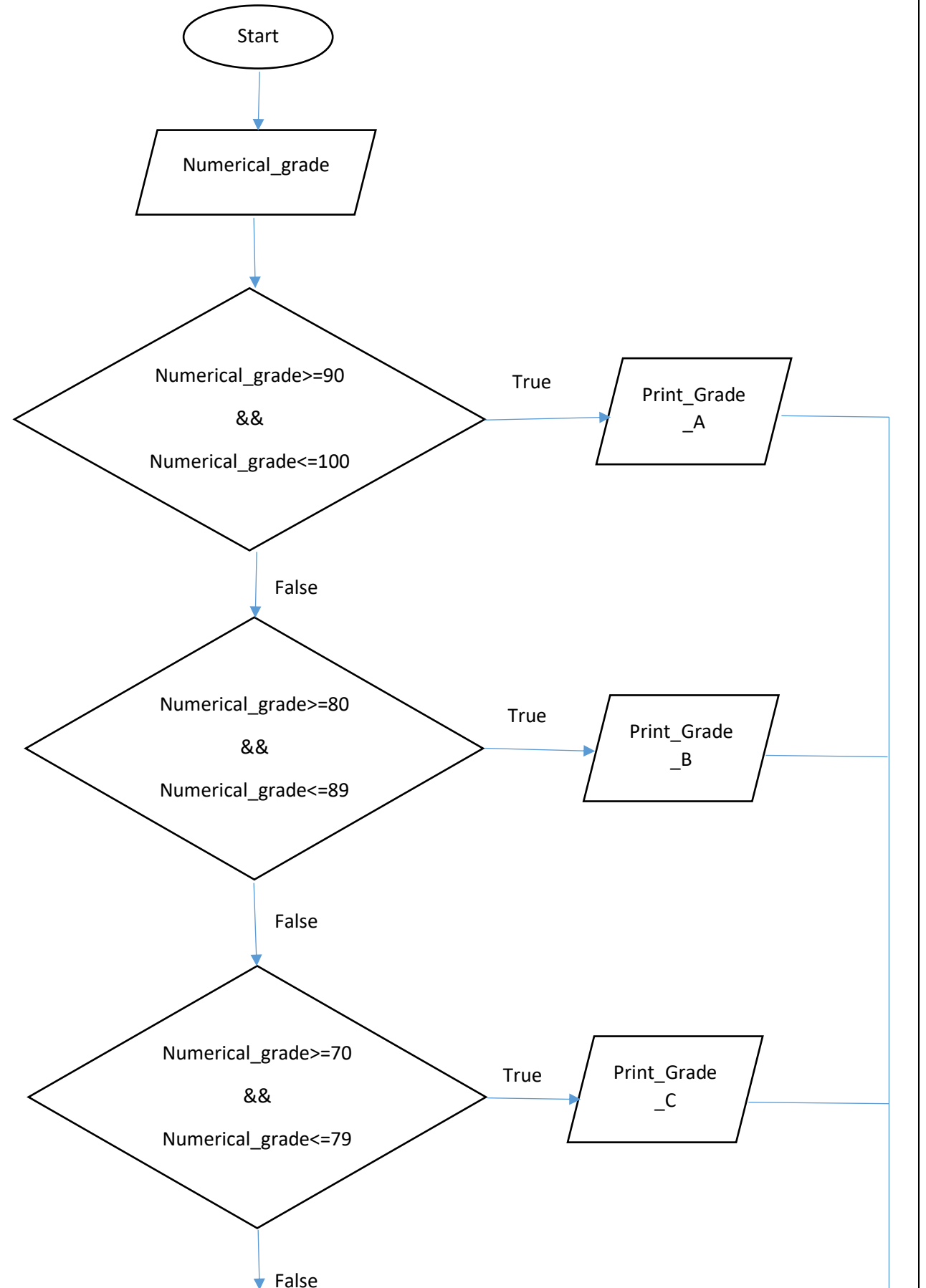
else

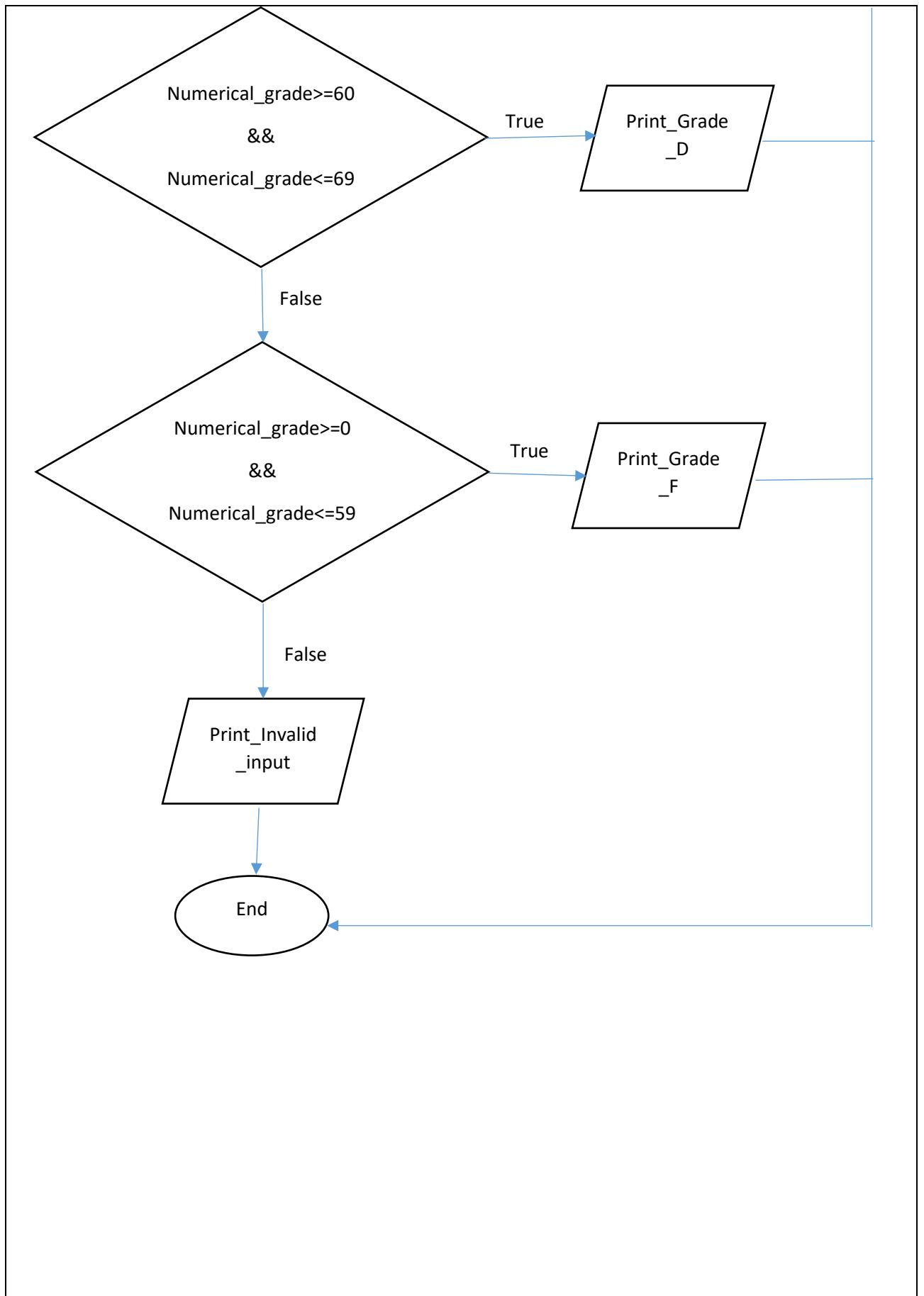
 Print_Invalid_input;

Print_Grade

End

FLOWCHART:





C++ PROGRAM:

```
#include <iostream>

using namespace std;

int main()
{
    int Numerical_grade;

    cout<<"What is your numerical grade ?\n=";
    cin>>Numerical_grade;

    if (Numerical_grade>=90 && Numerical_grade<=100)
        cout<<"\nGrade A";
    else if (Numerical_grade>=80 && Numerical_grade<=89)
        cout<<"\nGrade B";
    else if (Numerical_grade>=70 && Numerical_grade<=79)
        cout<<"\nGrade C";
    else if (Numerical_grade>=60 && Numerical_grade<=69)
        cout<<"\nGrade D";
    else if (Numerical_grade>=0 && Numerical_grade<=59)
        cout<<"\nGrade F";
    else
        cout<<"\nInvalid input";

    return 0;
}
```